ACADEMIC DISHONESTY POLICY

Academic honesty is one of the foundations of the educational mission and Catholic commitment of this University. Academic dishonesty, including such practices as cheating, plagiarism and fabrication, undermines the learning experience, and, as it involves fraud and deceit, is corrosive of the intellectual principles and is inconsistent with the ethical standards of this University. Academic dishonesty damages the sense of trust and community among students, faculty and administrators.

Types of Academic Dishonesty

Plagiarism is the act of presenting the work or methodology of another as if it were one's own. It includes quoting, paraphrasing, summarizing or utilizing the published work of others without proper acknowledgment, and, where appropriate, quotation marks. Improper use of one's own work is the unauthorized act of submitting work for a course that includes work done for previous courses and/or projects as though the work in question were newly done for the present course/ project. Fabrication is the act of artificially contriving or making up material, data or other information and submitting this as fact. Cheating is the act of deceiving, which includes such acts as receiving or communicating or receiving information from another during an examination, looking at another's examination (during the exam), using notes when prohibited during examinations, using electronic equipment to receive or communicate information during examinations, using any unauthorized electronic equipment during examinations, obtaining information about the questions or answers for an examination prior to the administering of the examination or whatever else is deemed contrary to the rules of fairness, including special rules designated by the professor in the course.

By Signing below, I verify that I have taken this test honestly and have neither cheated nor helped anyone else cheat; this is a mark of academic integrity.

Student Name (Please Print): CARLOS VERO (SOLUTIONS)

Student Signature: _____ Date: Oct 16, 2024

Student ID #: _____ Course Title/Number: _____

This test consists of four problems. You are required to work on any **three problems.** For full credit, you must **show all your work;** the answers alone are not sufficient. When you have finished the solution to a problem, **DRAW A BOX AROUND YOUR FINAL ANSWER.**

Numerical answers *must* have proper units. **Points will be subtracted for numerical answers with the wrong units or no units.** Use **scientific notation** for large or small numbers and give answers with **3 significant figures.**

Please make your work as legible as possible; partial credit for incorrect answers can only be assigned if your solution can be read. If you continue a solution to another page, please indicate where the solution is continued.

Indicate which problems you wish to have graded by circling *three* numbers below. **DO NOT** circle more than *three* numbers.

1 **2** **3** **4**

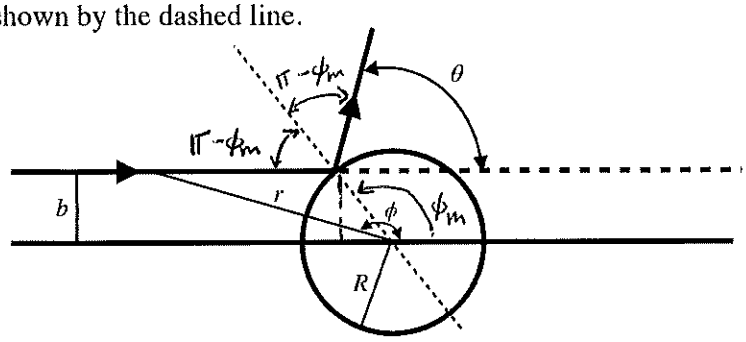
Make sure your name is on the test before handing it in.

1) (25 pts) The classical potential energy of a particle scattering from an infinitely massive hard sphere of radius R is given by

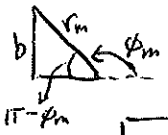
$$V(r) = \begin{cases} \infty, & r > R \\ 0, & r < R \end{cases}$$

where the potential has a finite non-zero value only at the surface, indicating that any classical interaction can only occur via a direct collision with the sphere.

For an impact parameter $b > R$, clearly there is no collision. However, for $b < R$, the particle scatters off the hard sphere at a deflection angle θ. The scattering is *symmetric* about the point of impact, as shown by the dashed line.



a) (5 pts) Express the distance of closest approach, r_m, in terms of the corresponding angle ϕ_m and impact parameter b (**Hint:** Can be obtained from simple geometry.)



$r_m \cdot \sin(\pi - \phi_m) = b$

$\sin(-(\phi_m - \pi)) = -\sin(\phi_m - \pi) = \sin\phi_m$

$$\boxed{r_m = \frac{b}{\sin\phi_m}} \quad \text{where } r_m = R$$

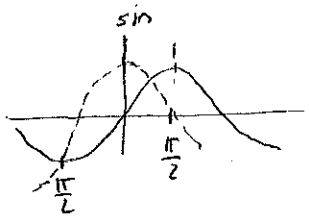
b) (5 pts) Using the result from part a), show that the scattering angle as a function of impact parameter b for $b < R$ is given by

$$\theta = 2\cos^{-1}(b/R)$$

Hint: Use the geometry at the distance of closest approach and the fact that the scattering is *symmetric* about the point of impact, to determine the angles about the dashed line (which should help translate $\phi_m \to \theta$)

$\pi = \theta + (\pi - \phi_m) + (\pi - \phi_m) = \theta + 2\pi - 2\phi_m$

$\Rightarrow \frac{2\phi_m}{2} = \frac{\theta}{2} + \frac{\pi}{2}$ $\quad \sin\phi_m = \sin\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \rightarrow$



$\sin\phi_m = \cos\frac{\theta}{2}$

$r_m = \frac{b}{\sin\phi_m} = \frac{b}{\cos\frac{\theta}{2}} = R \rightarrow \cos\frac{\theta}{2} = \frac{b}{R} \rightarrow \frac{\theta}{2} = \cos^{-1} b/R$

$\Rightarrow \boxed{\theta = 2\cos^{-1} b/R}$

c) (10 pts) Use the result from part b) and the general differential cross section formula, $d\sigma/d\Omega = (b/\sin\theta) \cdot db/d\theta$, to show

$$d\sigma/d\Omega = R^2/4$$

Hint: Use half-angle identities $\sin\theta/2 = \pm\sqrt{(1-\cos\theta)/2}$, $\cos\theta/2 = \pm\sqrt{(1+\cos\theta)/2}$ to simplify the problem, when necessary

From part b), $\theta = 2\cos^{-1} b/R$ or $b(\theta) = R\cos\frac{\theta}{2}$

$\frac{db}{d\theta} = -R\sin\left|\frac{\theta}{2}\right| \cdot \frac{1}{2}$ $\rightarrow$ $\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta}\left|\frac{db}{d\theta}\right| = \frac{R^2\cos\frac{\theta}{2}\sin\frac{\theta}{2}}{2\sin\theta}$

$\rightarrow \cos\frac{\theta}{2}\cdot\sin\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}\sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{(1-\cos^2\theta)}{4}} = \sqrt{\frac{\sin^2\theta}{4}} = \frac{\sin\theta}{2}$

$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{R^2}{2\sin\theta}\cdot\frac{\sin\theta}{2}$ $\Rightarrow$ $\boxed{\frac{d\sigma}{d\Omega} = \frac{R^2}{4}}$

d) (5 pts) Calculate that the total integrated cross section,

$$\sigma_{\text{total}} = \int (d\sigma/d\Omega)d\Omega$$

Hint: The differential solid angle of the sphere is $d\Omega = \sin\theta d\theta d\phi$ where $\theta : 0 \rightarrow \pi, \phi : 0 \rightarrow 2\pi$



$\sigma_{tot} = \int\left(\frac{d\sigma}{d\Omega}\right)d\Omega = \frac{R^2}{4}\int d\Omega$, $\int d\Omega = \iint \sin\theta\, d\theta\, d\phi$

$\rightarrow \int_0^{4\pi} d\phi = 2\pi$ $\int_0^{\pi}\sin\theta\, d\theta = -\cos\theta\Big|_0^{\pi} = -\cos(\pi) + \cos 0 = 2$

$\Rightarrow \int d\Omega = (2\pi)\cdot 2 = 4\pi$ $\Rightarrow$ $\sigma_{tot} = \frac{4\pi R^2}{4}$

$\boxed{\sigma_{tot} = \pi R^2}$

2) (25 pts) A boat with initial speed v_0 is launched on a lake. The boat is slowed by the water by a force $F = -\alpha e^{\beta v}$.

a) (5 pts) find an expression for the speed $v(t)$ $F = m\frac{dv}{dt} \rightarrow -\alpha e^{\beta v} = m\frac{dv}{dt}$

$$\rightarrow -\frac{\alpha}{m}\int_0^t dt = \int_{v_0}^{v} e^{-\beta v}\, dv$$

$$\rightarrow -\frac{1}{\beta} e^{-\beta v}\Big|_{v_0}^{v} = -\frac{1}{\beta}\left[e^{-\beta v} - e^{-\beta v_0}\right]$$

$$\Rightarrow -\frac{1}{\beta}\left[e^{-\beta v} - e^{-\beta v_0}\right] = -\frac{\alpha}{m}t \rightarrow \ln e^{-\beta v} = \ln\left[e^{-\beta v_0} + \frac{\alpha\beta}{m}t\right]$$

$$-\beta v = \ln\left(e^{-\beta v_0} + \frac{\alpha\beta}{m}t\right)$$

$$\rightarrow \boxed{v(t) = -\frac{1}{\beta}\ln\left(\frac{\alpha\beta}{m}t + e^{-\beta v_0}\right)}$$

b) (5 pts) find an expression of the time t_s it takes for the boat to stop

$$v(t = t_s) = -\frac{1}{\beta}\ln\left(\frac{\alpha\beta}{m}t_s + e^{-\beta v_0}\right) = 0$$

$$\Rightarrow \frac{\alpha\beta}{m}t_s + e^{-\beta v_0} = 1 \rightarrow \frac{\alpha\beta}{m}t_s = 1 - e^{-\beta v_0}$$

$$\boxed{t_s = \frac{m}{\alpha\beta}\left(1 - e^{-\beta v_0}\right)}$$

c) (**15 pts**) find an expression of the distance for the boat to stop

Note: helpful integral —> $\int \ln(at+b)dt = \frac{at+b}{a}\ln(at+b) - t$

$$\int_0^x dx = \int_0^{t_s} v(t)\,dt \rightarrow x = -\frac{1}{\beta}\int \ln\left(\frac{\alpha\beta}{m}t + e^{-\beta v_0}\right)dt$$

let $a \equiv \frac{\alpha\beta}{m}$, $b \equiv e^{-\beta v_0}$

$$= -\frac{1}{\beta}\int \ln(at+b)\,dt = -\frac{1}{\beta}\left[\frac{at+b}{a}\ln(at+b) - t\Big|_0^{t_s}\right]$$

$$= -\frac{1}{\beta}\left[\frac{at_s+b}{a}\ln(at_s+b) - t_s - \frac{b}{a}\ln b\right] \rightarrow \text{now substitute } a \text{ \& } b \text{ back and } t_s$$

$$at_s + b = \frac{\alpha\beta}{m}\underbrace{\left(\frac{m}{\alpha\beta}\cdot\left(1 - e^{-\beta v_0}\right)\right)}_{=t_s} + e^{-\beta v_0} = 1$$

$$= -\frac{1}{\beta}\left[\frac{1}{a}\underbrace{\ln(1)}_{\downarrow 0} - t_s - \frac{b}{a}\ln b\right]$$

$$= -\frac{1}{\beta}\left[-\frac{m}{\alpha\beta}\left(1 - e^{-\beta v_0}\right) - \frac{e^{-\beta v_0}\cdot m}{\alpha\beta}\ln\left(e^{-\beta v_0}\right)\right]$$

$$= -\frac{1}{\beta}\left(-\frac{m}{\alpha\beta}\right)\Big[\underbrace{1 - e^{-\beta v_0} + e^{-\beta v_0}(-\beta v_0)}_{1 - e^{-\beta v_0}(1+\beta v_0)}\Big]$$

$$\boxed{x(t_s) = \frac{m}{\alpha\beta^2}\left[1 - e^{-\beta v_0}(1+\beta v_0)\right]}$$

3) (25 pts) A particle moves in a logarithmic spiral orbit given by $r = ke^{\alpha\phi}$, where k, α are positive constants.

useful formulas: $E = \frac{1}{2}\mu\dot{r}^2 + \frac{\ell^2}{2\mu r^2} + V(r)$, $\ell = \mu r^2\dot{\phi}$, $\dot{r} = dr/d\phi \times \dot{\phi}$,
where $\dot{r} \equiv dr/dt$, $\dot{\phi} \equiv d\phi/dt$

a) (10 pts) Find a general expression for the potential, $V(r)$

$$V(r) = E - \frac{1}{2}\mu\left(\frac{dr}{dt}\right)^2 - \frac{\ell^2}{2\mu r^2} \rightarrow \frac{dr}{dt} = \frac{dr}{d\phi}\frac{d\phi}{dt}, \quad \frac{dr}{d\phi} = k\alpha e^{\alpha\phi} = \alpha r$$

$$\rightarrow \frac{dr}{dt} = (\alpha r)\left(\frac{\ell}{\mu r^2}\right) = \frac{\alpha\ell}{\mu r} \qquad \frac{d\phi}{dt} = \frac{\ell}{\mu r^2}$$

$$V(r) = E - \frac{1}{2}\mu\frac{\alpha^2\ell^2}{\mu^2 r^2} - \frac{\ell^2}{2\mu r^2} = E - \alpha^2\frac{\ell^2}{2\mu r^2} - \frac{\ell^2}{2\mu r^2}$$

$$\boxed{V(r) = E - \frac{\ell^2}{2\mu r^2}(\alpha^2 + 1)}$$

b) (5 pts) Use the result of part a) to show that the force law of this orbit is given by:

$F(r) = \frac{\ell^2}{\mu r^3}(1+\alpha^2)$ $\qquad F = -\nabla V(r) = -\frac{dV}{dr}$

$$\frac{dV}{dr} = +\frac{2\ell^2}{2\mu r^3}(\alpha^2+1)$$

$$F = -\frac{dV}{dr} \rightarrow \boxed{F = -\frac{\ell^2}{\mu r^3}(1+\alpha^2)}$$

c) (5 pts) determine a general expression for $\phi(t)$

Recall: $\frac{d\phi}{dt} = \frac{l}{\mu r^2}$ where $r(\phi) = ke^{\alpha\phi}$

$$\frac{d\phi}{dt} = l / (\mu k^2 e^{2\alpha\phi})$$

$$\int e^{+2\alpha\phi} d\phi = \int \frac{l}{\mu k^2} dt$$

$$+\frac{e^{+2\alpha\phi}}{2\alpha} + C = \frac{l}{\mu k^2} t \rightarrow \ln e^{+2\alpha\phi} = \ln\left(+\frac{2\alpha l}{\mu k^2} t + C'\right)$$

$$+2\alpha\phi = \ln\left(\frac{2\alpha l}{\mu k^2} t + C'\right)$$

$$\Rightarrow \boxed{\phi(t) = \frac{1}{2\alpha} \ln\left(\frac{2\alpha l}{\mu k^2} t + C'\right)}$$

d) (5 pts) determine a general expression for $r(t)$

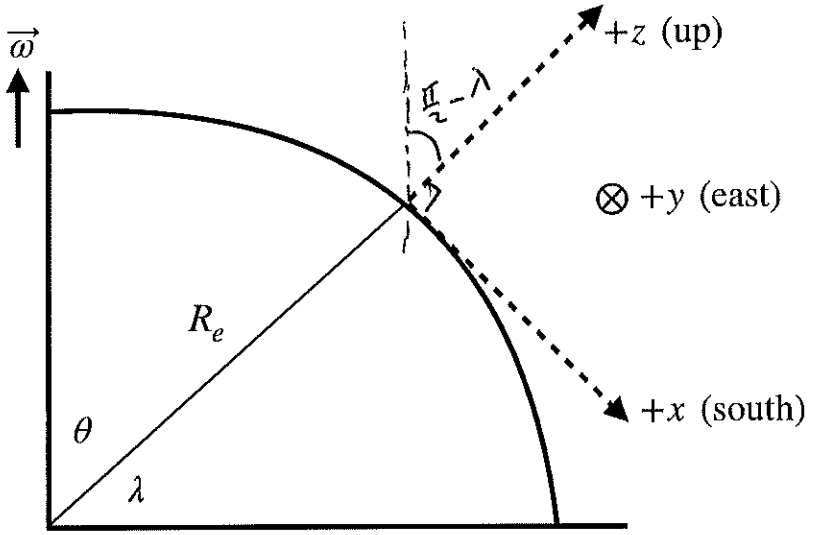
$r(\phi) = k e^{\alpha\phi}$ → substitute $\phi(t)$ into $r(\phi)$ → $r(t)$

$$r(t) = k e^{\alpha \cdot \frac{1}{2\alpha} \ln\left(\frac{2\alpha l}{\mu k^2} t + C'\right)^{1/2}}$$

$$r(t) = \left[k^2\left(\frac{2\alpha l}{\mu k^2} t + C'\right)^{1/2}\right] \Rightarrow \boxed{r(t) = \sqrt{\frac{2\alpha l}{\mu} t + k^2 C'}}$$

$\Rightarrow$ [illegible]

4) (25 pts) Shot towers were popular in the eighteen and nineteenth centuries for dropping melted lead down tall towers to form spheres for bullets. The lead solidified while falling and often landed in water to cool the lead bullets. Many such shot towers were built in New York State.



Use the local coordinate system as reference to solve this problem. Ignore second order effects due to Earth's angular speed, since $\omega_e^2 \sim 0$ (i.e., Only consider the acceleration terms to be gravity and Coriolis, $\vec{a}_{\text{tot}} = \vec{g}_o + \vec{a}_{\text{cor}}$ and assume small deflections due to Coriolis term, $v_x = v_y \approx 0$, when computing the Coriolis acceleration. The general expression for Newton's law in an accelerated frame as seen by an Earth observer is given by:

$$m\left(\frac{d^2\mathbf{r}}{dt^2}\right)_e = \mathbf{F}_g + \mathbf{F}' - 2m\boldsymbol{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_e - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

a) (5 pts) Determine the angular velocity vector of Earth in the localized coordinates in terms of the latitude angle, λ

Hint: In general, $\vec{\omega} = \omega_x\hat{x} + \omega_y\hat{y} + \omega_z\hat{z}$, where the components are projections of $\vec{\omega}$ onto each of the local coordinate axes.

$\omega_x = \vec{\omega} \cdot \hat{x} = \omega \cos(\pi - \lambda) = -\omega \cos\lambda$

$\omega_y = \vec{\omega} \cdot \hat{y} = \omega \cos(\frac{\pi}{2}) = 0$

$\omega_z = \vec{\omega} \cdot \hat{z} = \omega \cos(\frac{\pi}{2} - \lambda) = \omega \sin\lambda$

$$\boxed{\vec{\omega} = -\omega\cos\lambda\,\hat{x} + \omega\sin\lambda\,\hat{z}}$$

b) (5 pts) Calculate the magnitude and direction (*east, south, up?*) of the Coriolis acceleration, $\vec{a}_{cor}$, experienced by the falling projectile.

Hint: To obtain the final result, you will first need to find a general expression for the velocity of the projectile due to gravity, $v_z(t)$

$\vec{a}_{cor} = -2\vec{\omega} \times \vec{V}$ $\quad \vec{V} = V_z \hat{z}$ $\quad V_x = V_y \approx 0$ (small deflections)

$$\vec{\omega} \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega_x & 0 & \omega_z \\ 0 & 0 & V_z \end{vmatrix} = -\hat{y}\,\omega_x V_z \Rightarrow \vec{a}_{cor} = 2\omega_x V_z \hat{y} \text{ (east)}$$

determine V_z:

+z↑

$$\int -g_0 \hat{z} \int_0^{V_z} dV_z = \int_0^t -g_0\, dt$$

$$V_z = -g_0 t$$

Recall: $\omega_x = -\omega \cos\lambda$

$$\Rightarrow \boxed{\vec{a}_{cor} = 2\omega g_0 t \cos\lambda \hat{y}} \quad \text{"east deflection"}$$

c) (10 pts) Using the result obtained in part b), show that the general expression for the deflection of the projectile after falling back to Earth is given by:

$$\frac{1}{3}\omega g_o \cos\lambda \cdot (2z_o/g_o)^{3/2}$$

Hint: Note you will first need to find a general expression for the deflection as a function of time, and then evaluate it at $t = t_{fall}$, or the flight time of the projectile dropped from an initial height z_o

Recall: $a_{cor,y} = 2\omega g_o t \cos\lambda$

$$\int_0^{v_y} \frac{dv_y}{dt}dt = \int_0^t 2\omega g_o t\cos\lambda\, dt$$

$$v_y = \omega g_o t^2 \cos\lambda \rightarrow \int_0^y \frac{dy}{dt}dt = \int_0^t \omega g_o t^2 \cos\lambda\, dt$$

$$y(t) = \frac{1}{3}\omega g_o t^3 \cos\lambda$$

determine t_{FALL}:

$-g_o\hat{k}$ $v_z(t) = -g_o t$

$$\int_{z_o}^{z} \frac{dz}{dt}dt = \int_0^t -g_o t\, dt$$

$$z(t) = z_o - \frac{g_o t^2}{2}$$

Now at $t = t_{FALL}$ $z = 0$

$$\Rightarrow z_o - \frac{g_o t_{FALL}^2}{2} = 0$$

$$t_{FALL} = \left(\frac{2z_o}{g_o}\right)^{1/2}$$

substitute $t = t_{FALL}$ in $y(t)$:

$$y(t_{FALL}) = \frac{1}{3}\omega g_o \left(\frac{2z_o}{g_o}\right)^{3/2}\cos\lambda$$

d) (5 pts) Assume a shot tower was constructed at latitude $\lambda = 42°$ north, and the lead fell a distance of 27 m. How far did the lead bullets land from the direct vertical?

The angular velocity of Earth is $\omega_e \approx 7.29 \times 10^{5}\ \mathrm{s}^{-1}$ and $g_o = 9.81\ \mathrm{m/s^2}$

$$Y = \frac{1}{3}\left(7.29\times10^{-5}\ s^{-1}\right)\left(9.81\ \frac{m}{s^2}\right)\left(\frac{2.27m}{9.81\ m/s^2}\right)^{3/2}\cos 42°$$

$$\boxed{Y \approx 2.29 \times 10^{-3}\ m}$$